

curriculum vitae

Computational Research Engineer with a Ph.D. in Computational & Experimental Mechanics and deep expertise in physics-informed modeling, geometry processing, and surrogate model development. Specialized in designing and implementing scalable Python tools and data pipelines that transform large-scale simulation data (CAD/CAE) into fast, accurate, engineering-ready predictions. Strong experience in algorithm development, dataset curation, ablation studies, optimization techniques, and bridging complex multi-physics with practical, production-grade engineering software.

Personalia

Surname: Ruybalid
Given names: Andre, Paul
Degrees: Ph.D. | MSc. | BSc.
City: Los Alamos, New Mexico, U.S.A.
Phone number: +1 719 210 1746 (personal mobile)
E-mail: andreruybalid@gmail.com
Date of birth: 26 October 1987
Place of Birth: Farmington, New Mexico, U.S.A.
Nationality: American/Dutch

Professional experience

2023 - Present

R&D Engineer, Los Alamos National Laboratory

Materials Science and Technology Division (MST-8) | Los Alamos, NM, U.S.A.

- ❖ Designed, developed, and maintained multiple generations of physics-informed surrogate models and custom Python tooling to bridge polycrystalline mechanics and high-fidelity simulations to practical engineering-scale predictions. Twice recognized as R&D 100 Award finalist.
- ❖ Implemented N-dimensional (ND) finite element meshing algorithms using hypercubes and higher-order extrusions for complex geometries in high-dimensional spaces, supporting advanced multi-physics and geometry processing workflows.
- ❖ Architected large-scale simulation pipelines and curated high-quality datasets from thousands of low-length-scale simulations to train accurate data-driven surrogate models for viscoplastic material behavior.
- ❖ Developed and contributed production code for material models in the MOOSE/BISON C++ framework, including implementation, verification, validation, and uncertainty quantification workflows.
- ❖ Built interactive Python tools for data analysis, visualization, ablation studies, and workflow automation, significantly improving efficiency and enabling cross-functional team members to leverage complex modeling capabilities.
- ❖ Translated complex multiphysics modeling results into actionable engineering insights through technical reports and stakeholder presentations.

2021 – 2023

Postdoctoral Researcher, Los Alamos National Laboratory

Materials Science and Technology Division (MST-8) | Los Alamos, NM, U.S.A.

- ❖ Developed the surrogate model (“LAROMance”) for polycrystalline nuclear reactor materials by designing large-scale simulation pipelines and curating datasets from thousands of low length-scale simulations.
- ❖ Implemented physics-informed, data-driven surrogate modeling techniques to capture microstructure-dependent (nuclear) material behavior.

2019 – 2020

Design Engineer, ASML

DUV Def & Metrology group of Defectivity Department | Eindhoven, The Netherlands

Developed and maintained complex numerical algorithms and custom MATLAB tooling for high-precision nanoscale defect detection and geometry analysis on DUV lithography reticles. Designed, validated, and optimized metrology pipelines through

large-scale data analysis and cleanroom experimentation, delivering production-ready solutions under tight deadlines in cross-functional and industrial environments.

2013 – 2019

PhD Researcher, Eindhoven University of Technology

*Computational and Experimental Mechanics | Prof. Dr. Marc Geers & Dr. Johan Hoefnagels.
Philips Research, Materials Innovation Institute | Eindhoven, The Netherlands*

- ❖ Designed and implemented an integrated experimental-numerical framework combining finite element modeling, image processing, and digital image correlation (IDIC) for full-field material characterization and data analysis.
- ❖ Developed custom data processing pipelines and inverse optimization algorithms to identify constitutive material parameters from complex, heterogeneous experimental datasets.

Education

Eindhoven University of Technology, the Netherlands

Department of Mechanical Engineering

Division: Computational and Experimental Mechanics

Research Group: Mechanics of Materials

2013 – 2019 **PhD. in Mechanical Engineering**

Title: "Identification of multi-layer interface systems: a full-field, micro-mechanical approach."

2013 – 2019 **Graduate School of Engineering Mechanics**

Advanced Mechanics Courses

2010 – 2013 **MSc. Mechanical Engineering**

Minor: Biomedical Engineering

2006 – 2010 **BSc. Mechanical Engineering**

2000 – 2006 **Research-oriented secondary education (VWO)**

St.-Janscollege, Hoensbroek, The Netherlands

Technical Skills

Programming	Python (NumPy, SciPy, Pandas, Matplotlib), MATLAB, C++ (MOOSE framework), Linux/bash scripting.
Scientific Computing	Physics-informed surrogate model development, multi-physics simulation, large-scale dataset curation & ablation studies, inverse optimization, uncertainty quantification, geometry processing.
Software Development	Custom CAE tool & pipeline development, workflow automation, Python-to-C++ integration, production-grade code, verification & validation, modular design.
Domain Expertise	Computational mechanics, CAE workflows, finite element analysis, continuum mechanics, experimental-numerical frameworks, digital image correlation, (nuclear) materials modeling, microelectronic materials.

Selected journal publications (peer-reviewed)

P. Robbe, A.P. Ruybalid, A. Hegde, C. Bonneville, H.N. Najm, L. Capolungo and C. Safta . A comparison of surrogate constitutive models for viscoplastic creep simulation of HT-9 steel. *Computational Materials Science*, 262, 114367 (2026). <https://doi.org/10.1016/j.commatsci.2025.114367>.

A.P. Ruybalid, A. Tallman, W. Wen, C. Matthews, L. Capolungo. Data-Driven Surrogate Modeling with Microstructure-Sensitivity of Viscoplastic Creep in Grade 91 Steel. *Integrating Materials and Manufacturing Innovation* 13, 895–914 (2024). <https://doi.org/10.1007/s40192-024-00377-z>

A.P. Ruybalid, O. van der Sluis, M.G.D. Geers and J.P.M. Hoefnagels. Full-field identification of mixed-mode adhesion properties in a flexible, multi-layer microelectronic material system. *Engineering Fracture Mechanics*, 226, pp. 106879 (2020), <https://doi.org/10.1016/j.engfracmech.2020.106879>.

A.P. Ruybalid, O. van der Sluis, M.G.D. Geers, J.P.M. Hoefnagels. An *In-Situ*, Micro-Mechanical Setup with Accurate, Tri-Axial, Piezoelectric Force Sensing and Positioning. *Exp Mech* 60, 713–725 (2020), <https://doi.org/10.1007/s11340-020-00583-8>.

A.P. Ruybalid, J.P.M. Hoefnagels, O. van der Sluis, M.P.F.H.L. van Maris and M.G.D. Geers. Mixed-mode cohesive zone parameters from integrated digital image correlation on micrographs only. *International Journal of Solids and Structures*, 156-157, pp. 179-193 (2019), <https://doi.org/10.1016/j.ijsolstr.2018.08.010>.